Pocket Pico HF Packet Receive

Oct 16, 2026 · @Gene

Summary

The Pocket Pico now receives 300 baud HF packet: 10 of 10 test packets decoded through an IC-718, even with the two radios 70 Hz apart in tuning.

The HF firmware (the `pocketpicohf` branch of pfiliberti/pico_tnc) already transmitted 300 baud packets that other stations could decode. Receive did not work at all. The fix was a new receive decoder plus a timing fix for TXDELAY. Transmit audio itself was not changed.

	Before	After
Transmit 300 baud	Worked	Works (shorter TXDELAY)
Receive 300 baud	0 packets decoded	10 of 10 on test
Station 70 Hz off tune	n/a	10 of 10
Station 150 Hz off tune	n/a	10 of 10
TXDELAY 55 actually sent	about 2.2 s	about 0.6 s

The problem

Only the transmit side of the HF branch had been converted to 300 baud; the receive side was still the original 1200 baud VHF (Bell 202) design.

- Bit timing was set for 1200 baud, four times too fast for HF packet.
- The demodulator compared the audio with a copy of itself delayed 446 microseconds. That delay is tuned to tell 1200 Hz from 2200 Hz, not 1600 Hz from 1800 Hz.
- The audio filters passed 900-2500 Hz (input) and 1200 Hz (after the demodulator), sized for 1200 baud.

Run on 100 computer-made HF test packets, this code decoded none.

What was changed for receive

Receive was rebuilt around a new decoder that measures how strong the 1600 Hz (mark) and 1800 Hz (space) tones are over each bit and picks the stronger one.

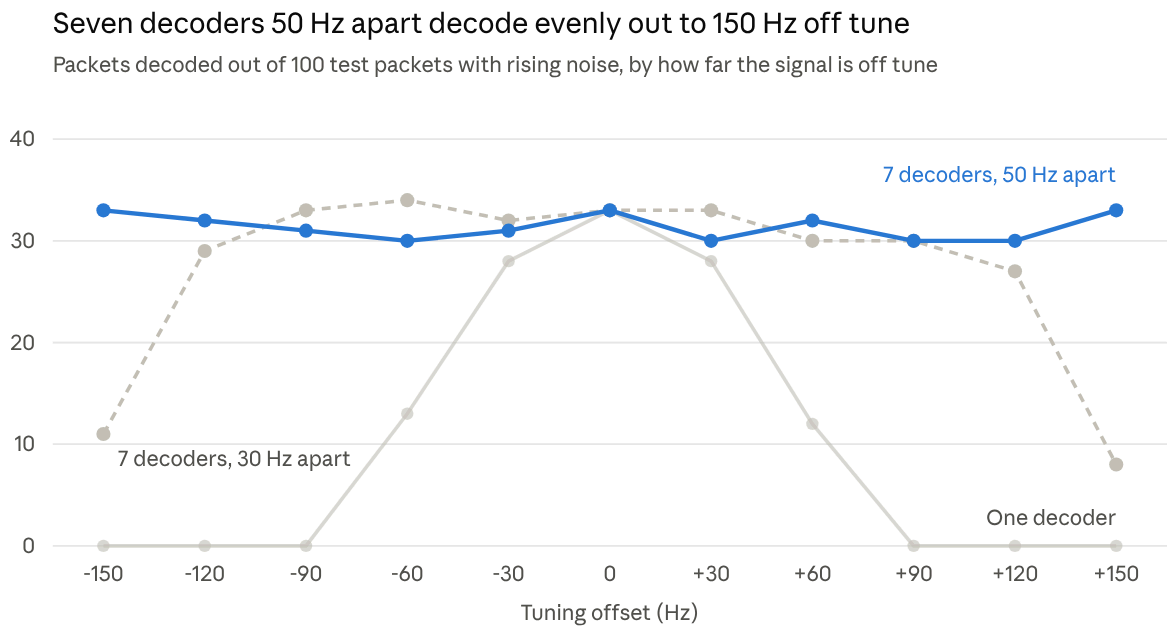
Setting	Old (1200 baud VHF)	New (300 baud HF)
Bit rate	1200 baud	300 baud
Audio samples per second	13,200	13,200 (unchanged)
Samples per bit	11	44
Input filter	900-2500 Hz, 25 taps	1100-2300 Hz, 31 taps
Tone detector	delay-and-multiply, 446 microseconds	mark/space tone energy over one bit
Decoders	1	7, tuned 50 Hz apart

Each of the seven decoders has its own bit clock (a PLL, as in Direwolf) and its own frame decoder. When any one of them gets a packet whose checksum (FCS) is good, the packet goes to the TNC. Copies of the same packet from the other decoders are dropped, so each packet shows once.

The new code is in two new files, `hf_demod.c` and `hf_demod.h`. Small changes hook it into `decode.c`, `tnc.c`, `tnc.h`, `receive.c` and `main.c`. The Pico keeps up easily: decoding takes at most 4 ms of every 10 ms of audio, and it uses about 13 KB more memory.

Why seven decoders

On the air, the IC-718 heard the FT-710 about 70 Hz low, and that was enough to make a single decoder drop packets. Two SSB radios rarely agree exactly on frequency, so HF stations are often 50-100 Hz off from each other.



PC bench test of the Pico decoder code · Direwolf gen_packets audio, 100 packets per offset, noise rising through each file

The seven decoders are tuned -150, -100, -50, 0, +50, +100 and +150 Hz from 1600/1800 Hz. Whichever one is closest to the station's real tones gets the packet. The first version used 30 Hz steps, like Direwolf's "7@30" setting; widening to 50 Hz steps keeps decoding even out to 180 Hz off tune, and costs at most about 10% for a signal halfway between two decoders. On the bench, Direwolf itself decoded 38 of 100 on tune, so the Pico is close to it.

TXDELAY fix

TXDELAY now means the same time at 300 baud as at 1200 baud: TXDELAY 55 sends about 0.6 s of sync flags before each packet, not 2.2 s.

TXDELAY (and TXTAIL) are set in 10 ms steps, and the firmware turns them into a number of flag bytes to send. That conversion assumed 1200 baud, where one flag takes 6.7 ms. At 300 baud a flag takes 27 ms, so every delay came out four times too long. The conversion in `send.c` now uses the actual baud rate. At 1200 baud nothing changes.

How it was tested

The final firmware decoded every test packet on the air and its transmissions were decoded by Direwolf.

Receive setup

1. A Yaesu FT-710 on 14.105 MHz, DATA-L, 10 W into a 50 ohm dummy load, played a sound file of 10 numbered 300 baud packets (1600/1800 Hz) through its USB audio.
2. The Icom IC-718 on 14.105 MHz LSB, on an outdoor 73 ft end-fed antenna, heard the dummy load's leakage at about S9+30.
3. The Pocket Pico, connected to the IC-718, printed the packets it decoded (`MONITOR ON`).
4. An OpenWebRX receiver with Direwolf decoded the same signal as a reference.

Results

Firmware / setup	Pocket Pico decoded
Original HF branch	0 of 10
One new decoder, radios 70 Hz apart	4 to 8 of 10
One new decoder, FT-710 retuned to match	10 of 10
Seven decoders 30 Hz apart, radios 70 Hz apart	10 of 10
Final firmware with TXDELAY fix	10 of 10
Seven decoders 50 Hz apart, radios about 150 Hz apart	10 of 10

Transmit: the Pocket Pico sent three packets through the IC-718 with TXDELAY 55, and OpenWebRX's Direwolf decoded all three.

Tip learned along the way: keep the transmit audio low. With the FT-710's ALC swinging up to +20, packets were distorted. With the audio turned down until ALC barely moved, Direwolf went from 9 to 10 of 10.

Getting it and installing it

The work is now part of the original project: download `pico_tnc.uf2` from the `pocketpicohf` branch of github.com/pfiliberti/pico_tnc (version 0.82 or later). All three changes are merged there: the receive decoders (pull request 10), the stuck-transmit fix (pull request 11) and the 50 Hz decoder spacing (pull request 13).

1. Unplug the Pico, hold its BOOTSEL button, and plug it into a computer's USB port.

2. A drive named RPI-RP2 appears. Copy `pico_tnc.uf2` onto it.
3. The Pico restarts by itself with the new firmware. A "Watch Dog Timer Failure" line right after flashing is normal: the Pico's loader restarts the chip that way.

Notes

- Set the radio to LSB. Mark/space order does not matter for packet, but both stations need to land in the same audio passband.
- With seven decoders, stations up to about 180 Hz off tune decode without retuning.
- Optional diagnostics for builders: uncomment `CODEX_RX_DIAGNOSTICS` in `tnc.h`, rebuild, and type `&` on the console. It shows which decoder caught each packet, the tone frequency heard, the audio level and timing.
- Built with the Raspberry Pi Pico SDK 2.2.0.

Credits: Pico TNC by JN1DFF; Pocket Pico hardware and HF branch by pfiliberti; HF receive and testing by Gene N7WDZ, with code written by Claude (Anthropic). The multi-decoder idea follows Direwolf by WB2OSZ.